

Hyunsik Jung

Data Scientist turning engineering and manufacturing data into operational ML/AI systems for quality and process decisions

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Summary

- Data Scientist who turns engineering and manufacturing data into operational ML/AI systems for quality and process decisions.
- Machine Learning Engineer / CL3 Staff Engineer at Samsung Electronics DX Division, with experience in TV manufacturing quality, miniLED optical simulation, device quality sensor logs, and operational risk models.
- Principal Data Scientist, AI Specialist, contributor to three A2-grade invention/patent disclosures, and technical interviewer for 20+ experienced-hire interviews.

Focus Areas

Manufacturing / Process Data	Analyze data with physical and engineering context, including process inspection data, quality logs, service history, optical signals, battery behavior, and device sensor telemetry.
Operational ML	Connect models to real workflows by designing thresholds, false-positive controls, review volume, retraining logic, dashboards, and handoff paths.
AI-ready Engineering Data	Build the semantic layer AI needs to use internal data correctly: marts, key definitions, domain rules, validation queries, Q-SQL pairs, and structured knowledge.

Core Skills

Programming / Tools	Python, SQL, BigQuery, PyQt, Streamlit, Django, Docker, Jupyter, Superset, Git/GitHub
ML / Statistics	XGBoost, Prophet/SARIMAX, regression, curve fitting, statistical significance testing, threshold design
Data / AI	Data mart design, JSON log parsing, semantic data layer, LLM knowledge preparation, Ontology/RDF/Knowledge Graph

Experience

Samsung Electronics, DX Division

Period	2018.09 - Present
Role	Machine Learning Engineer
Level	CL3 / Staff Engineer

Role History

2018.09 - 2020.12	AI & Big Data Lab, Visual Display Business. TV process latent defect prediction, miniLED BLU simulation, and service-parts demand forecasting.
2021.01 - 2025.10	Analytics Group, Big Data Center. Market-size forecasting, procurement cost analytics, DCS dashboard/mart, Smart Charging, eStore fraud, and SKU impact simulation.
2025.11 - Present	Gen AI Development Group, Big Data Center. DQA Data Agent, enterprise knowledge systems, Ontology/Knowledge Graph PoC, and AI Specialist work.

Representative Proof

DQA Data Agent / AI-ready Data	Prepared marts, key definitions, domain knowledge, Q-SQL pairs, and validation queries for quality sensor logs across 30+ device components, enabling hardware engineers to analyze logs in natural language.
Early-stage TV Process Defect Prediction	Connected process inspection data with global service-center defect/repair history to screen latent defects immediately after manufacturing. Designed threshold operation around aging-test capacity and line bottlenecks. Related defect rates dropped to roughly half.
BLU Simulator Development and Validation	Simulated miniLED BLU light distribution through curve fitting, regression, light-overlap validation, and Sobel-filter-based quality metrics. Delivered a PyQt Windows application with 90%+ average similarity to physical-panel images.
Galaxy Smart Charging	Developed upstream logic to infer individual sleep windows from device-level charging and app-usage logs, then switch long charging sessions to battery-protective slow charging. This became the Principal Data Scientist certification project and contributed to a product option.
eStore Trade-in Fraud Detection	Built an XGBoost model and operating thresholds/dashboard to detect trade-in incentive abuse at order time. Transferred to the U.S. sales subsidiary for operation, with 70% operational precision.

Certification / Recognition

Principal Data Scientist	Samsung DX Division's top internal Data Science Level 4 certification. Completed the 2023 one-year program and was certified in January 2024.
AI Specialist	Internal AI expert Level 3. Built an Expert-like Thinking Agent PoC that traces DQA error signals to candidate keys using electrical/electronic/physics knowledge, DQA specifications, and statistical validation.
Inventions / Patents	Contributor to three A2-grade invention/patent disclosures: process latent defect prediction, BLU light-distribution simulation, and service-parts demand forecasting with inventory anomaly detection.
Internal leadership	Served as a technical interviewer for 20+ experienced-hire interviews. Built and operated an internal Data Science learning platform using Django, Docker, and Jupyter.

Education

- KAIST, Ph.D. in Finance / Financial Engineering | 2018.08
- KAIST, B.S. in Mathematics and Industrial Engineering | early graduation in 3 years
- Korea Science Academy